

Overview: Goals of the Lab

This lab introduces the basics of logic circuit through 74LS00 TTL chip. Students will learn how to read a chip's configuration, functions, electrical characteristics and operating conditions in the manufacturer's datasheet. The lab also includes an introduction to SimUAid, a simulation software for logic circuits.

Step-by-Step Procedures and Results

Part 1: Getting Familiarized with the 74LS00 TTL Chip

74LS00 is a 14-pin device. It is a member of logic gate series of digital electronics. It is a TTL (Transistor to Transistor Logic). It represents _____ logic gate. The IC has four ____ gates hence it is called “_____ gate”. It has total ____ input and ____ output pins



Figure 1: Image of a SN74LS00N chip

1. Noticed that there are five fields in part number SN-74-LS-00-N. What do they indicate about this chip?

SN indicates ...

74 indicates ...

LS indicates ...

00 indicates ...

N indicates ...

- Using the datasheet, how many inputs to outputs are there in this TTL chip?

Your answer....

- Using the datasheet, what are pins 7 and 14 used for?

Your answer....

Link to the datasheet: <https://www.ti.com>

Part 2: TTL Characteristics of the 74LS00 TTL Chip

Table 1: Characteristics of 74LS00 TTL Chip

$V_{OH\ min}$	[V]
$V_{IH\ min}$	[V]
$V_{IL\ max}$	[V]
$V_{OL\ max}$	[V]
V_{CC}	[V]
$I_{OL\ max}$	[A]
$I_{OH\ max}$	[A]
$I_{IL\ max}$	[A]
$I_{IH\ max}$	[A]

- Explain what Fan-in and Fan-out means and how it is used in digital logic design. Then calculate the Fan-in and Fan-out of the 74LS00 TTL chip. What does the calculations imply?

Fan-in is

Fan-out is ...

$$\text{Fan-in} = \frac{A}{B} = x$$

$$\text{Fan-out} = \frac{C}{D} = y$$

For fan-in, the calculation implies ...

For fan-out, the calculation implies ...

5. Explain what Pull-up and Pull-down resistors mean and how it is used in digital logic design. Then calculate the Pull-up and Pull-down resistors for the 74LS00 TTL chip. What are the implications of these ranges?

Pull-up and Pull-down resistors are ...

For 74LS00 TTL chip:

Pull-up resistor: __[Ω] (show your calculations)

The value of R indicates ...

Pull-down resistor: __[Ω] (show your calculations)

The value of R indicates ...

6. What will happen if you do not include Pull-up or Pull-down resistors to a circuit using TTL chips?

Your answer...

7. What does positive logic and negative logic stand for? Could these have a connection with the Pull-up and Pull-down resistors implementation in logic circuits?

Your answer...

Part 3: Software Implementation Using SimUAid

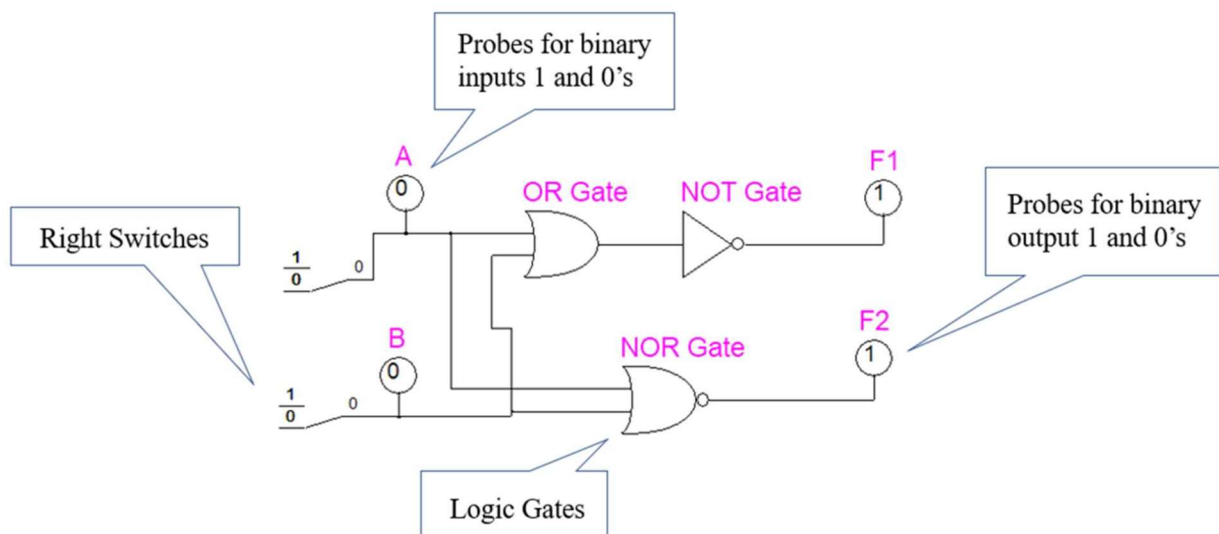


Figure 2: Logic Circuit of 74LS00 TTL Chip

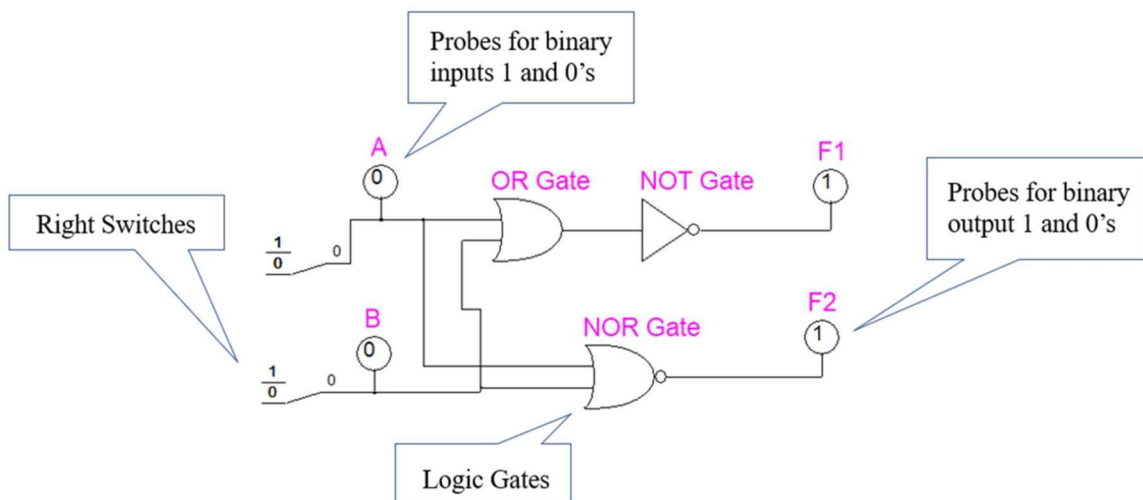


Figure 3: Equivalent Circuit of 74LS00 TTL Chip

Table 2: Truth Table for 74LS00 TTL Chip

Inputs		Output
A	B	Y
1	1	
1	0	
0	1	
0	0	

Conclusion

Through this lab, I have learned how to read a datasheet of a TTL chip. From there, I know its function and design, such as its logic circuit, the location of its input, output, ground and power supply pins. From the electrical characteristics and operating conditions, I can find its noise margin, further understand how the chip translate electrical currents into Boolean constants. I also learned how to use SimUAid to create a simulation of logic circuits and analyze its functions.